# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining**

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**COURSE OUTCOME COVERED**

**CO5:** Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

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**QUESTIONS: 05 Total Marks:50**

Annexure A

Constraint-Based Problem Solving

| ***Criteria*** | Problem Understanding ***(2.5)*** | Constraint Analysis ***(2.5)*** | Solution Development ***(2.5)*** | Application of Concepts ***(1.5)*** | Justification ***(1)*** | ***Total(10)*** |
|---|---|---|---|---|---|---|
| *Weightage* | *25%* | 25% | 25% | 15% | 10% | *100%* |
| *Q1* | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| *Q2* | | | | | | |
| *Q3* | | | | | | |
| *Q4* | | | | | | |
| *Q5* | | | | | | |

***Code of***

***Conduct:***

*I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.* ***Signature***

***of the student:***

**RUBRICS FOR CALCULATION:**

## Constraint-Based Problem Solving Assessment Rubric

| **Criteria** | **Weight ag e** | **Level 4 (Exempla ry )** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** |
|---|---|---|---|---|---|
| Problem Understan din g | 25% | Clearly defines the problem and identifies all relevant constraints accurately | Identifies most constraints with minor gaps | Partial understandi ng; misses key constraints | Fails to identify constraints correctly |
| Constraint Analysis | 25% | Analyzes constraints effectively and prioritizes them logically | Provides reasonable analysis with some prioritization | Limited analysis; weak prioritization | No meaningful analysis or prioritization |

| | | | | | |
|---|---|---|---|---|---|
| Solution Development | 25% | Develops optimal and feasible solutions satisfying all constraints | Develops workable solutions with minor limitations | Solutions partially satisfy constraints | Solutions do not meet constraints |
| Application of Concepts | 15% | Effectively applies appropriate concepts, methods, or tools | Applies concepts with minor errors | Limited application; requires guidance | Unable to apply relevant concepts |
| Justification | 10% | Provides strong justification for decisions with logical reasoning | Provides justification with some clarity | Weak or unclear justification | No justification provided |

SIMATS ENGINEERING
Assessment Tool 1 — Constraint-Based Problem Solving

# Association Rule Mining — Model Answer Key CSA16 · Data Warehousing and Data Mining • CO5

| | | | |
|---|---|---|---|
| **Course** | CSA16 — Data Warehousing & Data Mining | **CO5** | Association rule mining |
| **Questions** | 05 | **Total** | 50 Marks · Weightage 35% |

**CO5 Statement.** Apply association-rule-mining techniques to discover interesting relationships and patterns within large datasets.

*Definitions used throughout:* ***support(X) = (transactions containing X) / (total transactions); confidence(A→B) = support(A ∪ B) / support(A)****. Each answer states the given data and constraints, shows the support/confidence working, and justifies the result against the constraints — matching the rubric (Problem Understanding, Constraint Analysis, Solution Development, Application of Concepts, Justification).*

*Where a question omits a threshold, a reasonable value is assumed and stated explicitly.*

**Given & constraints**

- Transactions (N=5): P1{Fever,Cough}, P2{Fever,Headache}, P3{Cough,Headache}, P4{Fever,Cough,Headache}, P5{Fever,Fatigue}
- Min support = 50% (count ≥ 3 of 5) • Min confidence = 70%
- Constraint: every generated rule must include the item **Fever**

## Step 1 — item (1-itemset) support

| | | | | |
|---|---|---|---|---|
| **Fever** | P1,P2,P4,P5 | 4 | 80% | **Yes** |
| **Cough** | P1,P3,P4 | 3 | 60% | **Yes** |
| **Headache** | P2,P3,P4 | 3 | 60% | **Yes** |
| **Fatigue** | P5 | 1 | 20% | **No** |

## Step 2 — Fever-containing pairs and candidate rules

| | | | | |
|---|---|---|---|---|
| **{Fever, Cough}** | 40% | **No** | — | — |
| **{Fever, Headache}** | 40% | **No** | — | — |
| **Fever → Cough** | 40% | **No** | 40/80 = 50% | **No** |
| **Cough → Fever** | 40% | **No** | 40/60 = 66.7% | **No** |
| **Fever → Headache** | 40% | **No** | 40/80 = 50% | **No** |
| **Headache → Fever** | 40% | **No** | 40/60 = 66.7% | **No** |

**Interpretation & justification**

No Fever-containing pair reaches 50% support (the strongest is only 40%), and even ignoring support, no rule reaches 70% confidence (best is 66.7%).

**Result: the set of valid rules is empty** — under this small dataset the three constraints (support 50%, confidence 70%, must-include Fever) cannot be satisfied together.

This is itself the correct, defensible finding: the closest candidates are Cough→Fever and Headache→Fever (66.7% confidence). Lowering support to 40% and confidence to ~65% would admit them.



**Given & constraints**

- Transactions (N=5): S1{Python,SQL}, S2{Python,DataScience}, S3{SQL,MachineLearning}, S4{Python,SQL,DataScience}, S5{Python,MachineLearning}
- Min support = 40% (count ≥ 2 of 5)
- Constraint: each reported itemset must contain **at least two courses**

## Step 1 — 1-itemset support

| | | | |
|---|---|---|---|
| **Python** | 4 | 80% | **Yes** |
| **SQL** | 3 | 60% | **Yes** |
| **Data Science** | 2 | 40% | **Yes** |
| **Machine Learning** | 2 | 40% | **Yes** |

## Step 2 — 2-itemsets (constraint: size ≥ 2)

| | | | | |
|---|---|---|---|---|
| **{Python, SQL}** | S1,S4 | 2 | 40% | **Yes** |
| **{Python, Data Science}** | S2,S4 | 2 | 40% | **Yes** |
| **{Python, Machine Learning}** | S5 | 1 | 20% | **No** |
| **{SQL, Machine Learning}** | S3 | 1 | 20% | **No** |

| | | | | |
|---|---|---|---|---|
| {SQL, Data Science} | S4 | 1 | 20% | No |
| {Data Science, Machine Learning} | — | 0 | 0% | No |

The only candidate 3-itemset {Python, SQL, Data Science} occurs in S4 only (support 20%) → not frequent. **Frequent itemsets meeting the constraint: {Python, SQL} and {Python, Data Science}.**

**Interpretation & justification**

The 'at least two courses' rule is a **length (anti-monotone) constraint**: single-course itemsets are excluded from the output, so mining focuses only on multi-course combinations.
Combined with Apriori's downward-closure (any superset of an infrequent set is infrequent), it prunes the candidate space early — fewer candidates are generated and support-counted, reducing mining cost.



**Given & constraints**

- Transactions (N=5): T1{DM,PP}, T2{DM,DBS}, T3{PP,DBS}, T4{DM,PP,DBS}, T5{DM,AI} (DM=Data Mining, PP=Python Prog., DBS=Database Systems, AI=Artificial Intelligence)
- Constraint: maximum itemset size = 3 • assumed min support = 40% (count ≥ 2)

## Step 1 — 1-itemsets (L1)

| | | | |
|---|---|---|---|
| Data Mining (DM) | 4 | 80% | Yes |
| Python Prog. (PP) | 3 | 60% | Yes |
| Database Systems (DBS) | 3 | 60% | Yes |
| Artificial Int. (AI) | 1 | 20% | No |

## Step 2 — 2-itemsets (L2) and 3-itemsets (L3)

| | | | | |
|---|---|---|---|---|
| {DM, PP} | T1,T4 | 2 | 40% | Yes |
| {DM, DBS} | T2,T4 | 2 | 40% | Yes |
| {PP, DBS} | T3,T4 | 2 | 40% | Yes |
| {DM, PP, DBS} | T4 | 1 | 20% | No |

The single 3-itemset candidate {DM, PP, DBS} appears only in T4 (20%) → infrequent. **Frequent itemsets: {DM}, {PP}, {DBS}, {DM,PP}, {DM,DBS}, {PP,DBS}.**

**Interpretation & justification**

Capping itemset size at 3 bounds candidate generation to sizes 1–3 (C(n,1)+C(n,2)+C(n,3)), preventing the combinatorial blow-up of 4-itemsets and beyond.

Fewer candidates → fewer database scans and support counts → lower computational complexity. Here the cap is not even reached, because {DM,PP,DBS} is infrequent and Apriori's pruning already stops growth.



**Given & constraints**

■ Transactions (N=5): S1{Eng,CS,ML}, S2{Eng,IT}, S3{Eng,CS}, S4{Eng,CS,ML}, S5{Eng,IT} ■ Hierarchy: Engineering → {Computer Science → {Machine Learning}, Information Technology} ■ Constraint: generate rules only at the **Computer Science (CS)** level • assumed support ≥ 40%, confidence ≥ 70%

## Step 1 — support by hierarchy level

| | | | |
|---|---|---|---|
| **Engineering** | Top (root) | 5 | 100% |
| **Computer Science (CS)** | Mid | 3 | 60% |
| **Information Technology** | Mid | 2 | 40% |
| **Machine Learning (ML)** | Leaf (under CS) | 2 | 40% |

## Step 2 — rules at the CS level

| | | | |
|---|---|---|---|
| **ML → CS** | 2/5 = 40% | 2/2 = 100% | **Yes** |
| **CS → ML** | 2/5 = 40% | 2/3 = 66.7% | **No (conf)** |

**Interpretation & justification**

At the CS level the strong rule **ML → CS (support 40%, confidence 100%)** holds: every student who takes Machine Learning also takes Computer Science.

**Hierarchy effect:** support rises toward the root — Engineering is 100% (trivial/too general), while leaf items (ML) have low support. Mining at the CS level filters out the over-general Engineering rules and yields specific, actionable associations. Deeper levels may need a lower (level-specific) minimum support so genuine patterns are not lost.

**Given & constraints**

■ Records (N=5) over dimensions {Age Group, Income Level, Policy}: C1(Young,High,Health), C2(Middle,Medium,Vehicle), C3(Young,Low,Travel), C4(Senior,High,Health), C5(Young,Medium,Vehicle) ■ Constraint: mine only where **Age Group = Young** (customers C1, C3, C5) — patterns of Income → Policy

### Step 1 — the 'Young' slice

| | | | |
|---|---|---|---|
| **C1** | Young | High | Health Insurance |
| **C3** | Young | Low | Travel Insurance |
| **C5** | Young | Medium | Vehicle Insurance |

### Step 2 — multidimensional rules (support over all 5 records)

| | | | |
|---|---|---|---|
| **Young ∧ Income=High → Health** | 1/5 = 20% | 100% | **Strong** |
| **Young ∧ Income=Medium → Vehicle** | 1/5 = 20% | 100% | **Strong** |
| **Young ∧ Income=Low → Travel** | 1/5 = 20% | 100% | **Strong** |

**Interpretation & justification**

Within the Young segment there is a clean one-to-one **income → policy** mapping: High→Health, Medium→Vehicle, Low→Travel — each at 100% confidence (though low support, as each covers one of three young customers).

**Targeted marketing:** offer health plans to young high-income customers, vehicle policies to young middle-income, and travel policies to young low-income. The Age=Young constraint slices the data so mining yields segment-specific, directly actionable cross-dimension patterns; a larger sample would be needed to confirm the support.